CPR+
Captive Portal Remediation Tool

CPR+ / WHITEWALKER CAPABILITY INVENTORY

Endpoint Network State Detection, Remediation, and Telemetry Framework

A truthful use-case breakdown of what CPR+ does, why it exists, and how the framework coordinates DHCP/unlock triggers, posture recovery, captive portal handling, VPN hairpin prevention, Haystack monitoring, logging, and self-update.

Prepared for architecture / operations review — Steve Horton · CPR_3.14_RC2.0.2

1. Executive summary

CPR+ / WhiteWalker is an endpoint-side network-state framework. It is launched by Windows Task Scheduler when a device receives a DHCP address or when the user unlocks the endpoint. The main entry point, `WW_main.ps1`, waits for the network stack to settle, classifies the endpoint's real network state, and then routes to the smallest reasonable action.

> **Plain-English definition:** CPR+ lets an endpoint answer the question, "Where am I, what network condition am I actually in, and what should I do next without making things worse?"

The framework covers more than captive portal remediation. It handles Cisco ISE posture redirect recovery, walled-garden portal detection, VPN state handling, on-

prem VPN hairpin prevention, trusted SSID tie-breaking, Haystack event monitoring, syslog queueing, local diagnostics, state tracking, and gated self-update.

2

Primary launch triggers: DHCP/address change and unlock

50+

Operational behaviors and remediation paths documented here

740+

Pester test cases (It blocks) across 26 test files; many are static source checks, and not all have been run on a live endpoint

Architecture flow

Primary use cases

Capability matrix

Haystack and telemetry

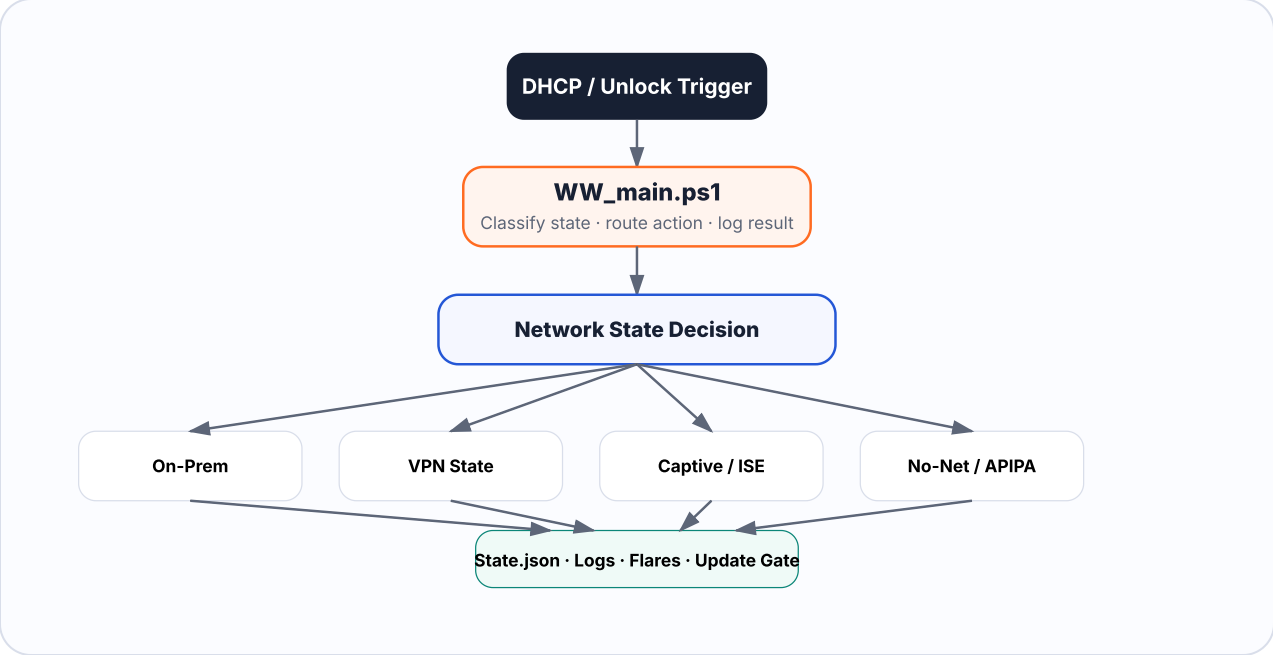
Safety controls

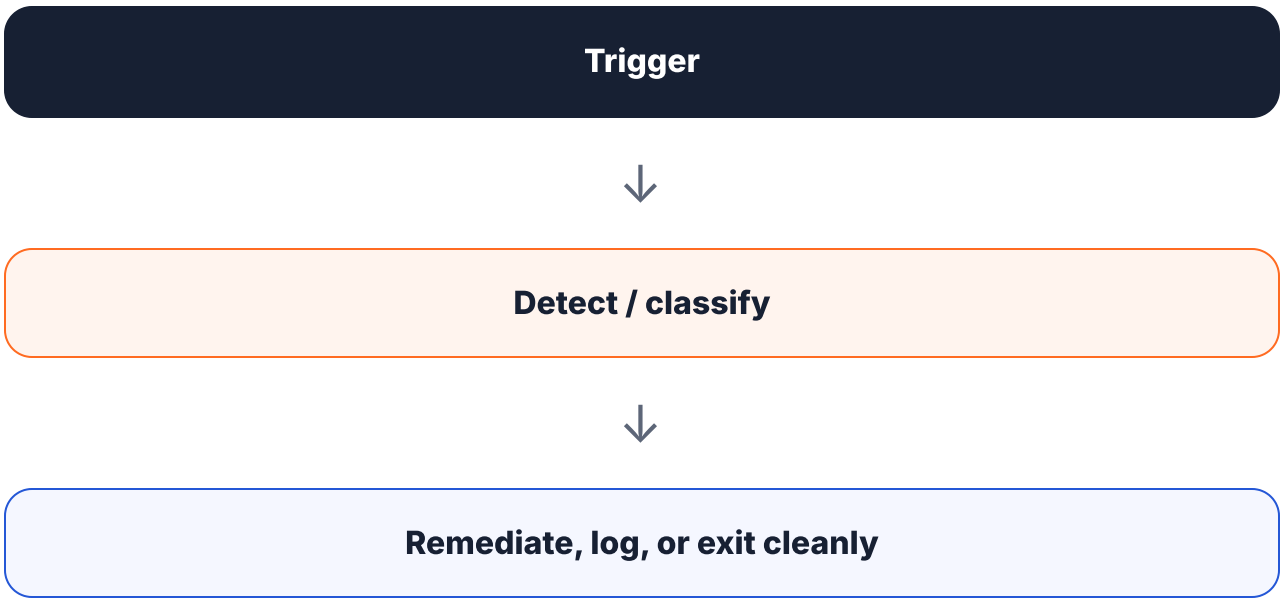
Summary list

Decision trees

2. Framework architecture

The root of the framework is `WW_main.ps1`. Supporting scripts own specific actions, so the main script can stay focused on classification, decision routing, state updates, and logging.





3. Primary use cases covered

Network access recovery

- ISE employee posture redirect detection
- Posture rescan trigger and compliance verification
- Captive portal browser launch
- Guest portal and walled-garden handling
- APIPA / no-network safe exit

VPN behavior control

- User tunnel and management tunnel detection
- VPN stuck/intermediate state handling
- SSID-change checks while VPN is active
- On-prem VPN hairpin prevention
- Hosts-file blackhole cleanup on non-on-prem states

Operations telemetry

- State file maintained for downstream scripts
- FlareGun event routing
- Haystack needle monitoring
- Syslog/Splunk forwarding with offline queue
- Diagnostic and tail helper scripts

---

4. Capability matrix

Area	What CPR+ does	Operational value
Early VPN gate	First decision of every run: if the VPN is stuck in Connecting/Reconnecting with no usable IP, after wake, or after an SSID jump, WhiteWalker disconnects it before probing. A fully Connected VPN is never touched.	Stops the client fighting the network change and keeps every later probe honest.
Critical path vs maintenance	Time-critical remediation runs first. Self-update, task sync and vault sync run afterward in a maintenance stage, only when online and the share is reachable.	Users get their network back first; housekeeping never delays recovery.
Bounded vpncli	Every vpncli call has a timeout and is killed if it hangs. After WhiteWalker cuts a VPN, a Connecting/Reconnecting bounce is re-disconnected a limited number of times.	A client that auto-reconnects on a captive network can no longer hang a run.
Captive portal runner modes	Base mode observes while Cisco native handling opens its browser. The full runner (private browser) is used only if escalation is enabled in configuration.	Lets the fleet choose observe-only or active browser launch per policy.
Tee (debug) levels	Levels 1-5 set by flag file; plain flags revert after 14 days, _keep flags stick. A 10-minute cooldown holds flag changes back. Master and active profile control which events log at each level.	Field debugging without a redeploy, and without leaving verbose logging on forever.
Combined update + tee flag	cpr_update_<dev|prod>_tee_<1-5>[_keep].flag installs the update and, only on success, sets the tee level.	One action to roll out a build and turn on debug logging for it.

Area	What CPR+ does	Operational value
Log write retry	Log writes retry on lock; if all retries fail the line goes to log_write_failures.txt.	Lines are no longer lost silently when a file is locked.
Admin console	Menu-driven tool for tee level, self-update triggers (with optional tee level), diagnostics and syslog path checks.	Support staff can operate WhiteWalker without hand-editing files.
DHCP / unlock execution	Runs automatically after address changes and unlock events, then waits briefly for the network stack to settle.	Catches the moment where most endpoint network failures show up without requiring the user to launch a tool.
State classification	Classifies on-prem, off-prem, user tunnel, management tunnel, captive portal, walled garden, untrusted SSID, and transient no-network states.	Prevents one-size-fits-all remediation. The endpoint acts based on what is actually happening.
ISE posture recovery	Detects posture redirects, triggers Cisco posture rescan, watches for compliance, and verifies post-compliance access.	Reduces tickets where the endpoint is connected but trapped behind posture ACLs.
Captive portal	Routes captive portal handling into the user context so browser and toast	Fixes hotel, guest, and venue

Area	What CPR+ does	Operational value
remediation	workflows happen where the user can interact.	networks without SYSTEM-context browser weirdness.
Walled-garden detection	Identifies networks where the gateway is reachable but DNS or HTTPS validation is blocked until portal acceptance.	Avoids mislabeling hotel-style captive portals as simple internet failure.
VPN hairpin prevention	When safely confirmed on-prem, sinkholes VPN headend names to localhost using a managed hosts-file block.	Prevents unnecessary VPN connection attempts from inside the corporate network.
Trusted SSID tie-breaker	Uses hardcoded and synced trusted SSID lists to avoid false on-prem detection from stale VPN routes or misleading DC reachability.	Prevents CPR+ from disconnecting or blocking VPN while the user is really off-prem.
Haystack monitoring	Watches for configured Windows Event Log needles and writes structured hits.	Adds lightweight endpoint event detection without a heavy always-polling agent loop.
Syslog queueing	Sends structured events to syslog/Splunk when reachable; queues locally when offline and drains later.	Keeps telemetry useful without breaking endpoint workflows when

Area	What CPR+ does	Operational value
		the collector is unreachable.
Self-update	Uses DEV/PROD flag files and network-state gating to pull updates from NETLOGON with cooldown controls.	Supports controlled repair and rollout without touching every endpoint manually.
Logging and diagnostics	Writes decision breadcrumbs, state changes, warnings, exit reasons, and rotating logs.	Gives IT teams evidence instead of guesses.

5. Detailed use case inventory

DHCP / unlock network state evaluation

CPR+ detects the active network context after DHCP and unlock events. It checks preflight behavior, redirect signals, adapter state, VPN state, SSID trust, and gateway/DNS conditions before deciding what to do.

on_prem user_tun mgmt_tun off_prem_no_vpn captive_portal no_net_transient

ISE posture rescan automation

When posture redirect behavior is detected, CPR+ routes into the posture workflow, triggers rescan, checks service availability, monitors compliance, and verifies that network access actually recovered.

Captive portal and guest Wi-Fi

Portal handling is intentionally moved into a user-context runner. That keeps the main script from launching browsers as SYSTEM and gives the user a usable auth window.

Walled-garden networks

CPR+ distinguishes DNS-blocking portal conditions from general network failure. Gateway reachable plus DNS failure can route to the portal browser instead of dumping into a dead-end no-network state.

VPN blackhole framework

When the device is confirmed on-prem, CPR+ can apply a controlled hosts-file sinkhole for configured VPN headends. Removal is deliberately broad and safety-biased on startup, shutdown, off-prem, VPN, captive, and no-network states.

Wake and SSID-change recovery

After wake or Wi-Fi movement, VPN can appear connected while blocking the local auth path. CPR+ checks SSID changes, gives VPN a grace window, and can disconnect only when it appears stuck or harmful.

---

6. Haystack, syslog, and telemetry

Haystack is the event-monitoring side of CPR+. It watches for configured “needle” patterns in Windows Event Logs and writes structured hits. Those hits can be sent to syslog/Splunk when the network state says the collector should be reachable.

Important distinction: Haystack is not the root decision engine. `WW_main.ps1` detects current network state and enables network-aware decisions. Haystack consumes that state so telemetry can be sent, cached, or deferred safely.

Haystack behavior	Why it matters
Merges local and shared needle files	Allows local testing before promoting a shared detection pattern.
Task Scheduler event wakeup	Avoids a constant polling loop.
Structured `NEEDLE_HIT` output	Makes downstream parsing sane.
Offline syslog cache	Prevents data loss when the endpoint is off-net or collector is unreachable.
FIFO drain with retention limits	Keeps queue behavior bounded and predictable.

7. Safety and “do no harm” controls

CPR+ is built around conservative cleanup and cooldown behavior. The general pattern is: detect carefully, act narrowly, log clearly, and remove risky state aggressively when conditions change.

- Blackhole removal on startup
- Blackhole removal on all non-on-prem states
- Shutdown task to clear VPN sinkholes before reboot
- Trusted SSID tiebreaker before on-prem actions
- Cooldowns for flares, rescans, updates, and syncs

- Local cache fallback when NETLOGON is unreachable
- Version and hash checks before config replacement
- Size-based log rotation
- Whatlf, debug, and trace modes
- APIPA/no-network early exits
- Portal and browser handling isolated into user context
- Self-update gated by current network state

---

8. Test and validation coverage

The RC2.0.2 tree contains about 748 Pester `It` blocks in 26 files. A portion are behavior tests that load the real functions by AST; many are static source checks. The last full Pester run on a live endpoint (v5.3.3, June 2026) was against an older build, so treat the newer RC2 tests as written but not yet run on-box.

~534

tests in tests/ (RC2 contract, recovery, build identity, admin console)

~155

captive portal runner tests (full + base runner)

Tested area	Examples
Main logic	Wake detection, flare behavior, state management, preflight classification, VPN status parsing.
Captive portal runner	Browser launch, PID tracking, smart wait polling, validation handling, Cisco browser cleanup.

Tested area	Examples
Blackhole behavior	Add/remove paths, disabled behavior, startup safety removal.
Diagnostics	VPN status parsing and diagnostic collection paths.

9. Summary list: what CPR+ does

1. Runs on DHCP/address change and unlock.
2. Classifies current endpoint network state.
3. Detects clean network access through preflight behavior.
4. Detects ISE posture redirect conditions.
5. Triggers Cisco posture rescan.
6. Verifies compliance and post-compliance access.
7. Detects standard captive portals.
8. Detects ISE guest portals.
9. Detects DNS-blocking walled gardens.
10. Launches portal remediation in user context.
11. Tracks portal remediation state.
12. Handles portal failure and timeout paths.
13. Detects user VPN tunnel state.
14. Detects management VPN tunnel state.
15. Recognizes VPN intermediate/stuck states.
16. Handles SSID changes while VPN is active.
17. Disconnects VPN only when it appears to block local auth.
18. Prevents on-prem VPN hairpinning.
19. Applies managed hosts-file sinkholes for VPN headends.
20. Removes sinkholes on startup, shutdown, and non-on-prem states.

21. Supports AD allow-group exemptions for blackhole behavior.
22. Syncs blackhole configuration from NETLOGON.
23. Uses trusted SSID tie-breaking to avoid false on-prem actions.
24. Syncs trusted SSID config with local fallback.
25. Detects APIPA and no-usable-network conditions.
26. Waits for corporate Wi-Fi address assignment when appropriate.
27. Logs MAC randomization state and warnings.
28. Cleans up interfering Cisco browser/helper processes.
29. Fires structured FlareGun events.
30. Maintains `state.json` for downstream scripts.
31. Runs Haystack event monitoring when enabled.
32. Matches Event Log needles from config.
33. Writes structured `NEEDLE_HIT` records.
34. Forwards hits to syslog/Splunk.
35. Caches syslog hits offline and drains later.
36. Supports DEV/PROD self-update flags.
37. Gates updates by network state and share reachability.
38. Uses cooldowns to prevent loops and share thrash.
39. Rotates logs by size.
40. Provides useful local evidence for IT troubleshooting.

---

10. Expandable decision trees

The trees below are collapsed by default so the main report stays readable. They are intended for reviewers, support engineers, and the brave few who actually want to see how the sausage gets made.

> **Reading note:** These are architecture-level decision trees, not a line-by-line code dump. They preserve the intent of the CPR+ flows while avoiding every

internal helper call and guard clause.

Main state engine

VPN gate and stages

ISE posture

Captive portal

VPN blackhole

Haystack

Syslog drain

Self-update

FlareGun

Safety cleanup

Trusted SSID

WW_main.ps1 — primary state classification +

ISE posture remediation decision tree +

Captive portal and walled-garden decision tree +

VPN hairpin / blackhole decision tree +

Trusted SSID tie-breaker decision tree +

Haystack needle handling decision tree +

Syslog / Splunk cache-and-drain decision tree +

Early VPN gate and run stages +

Self-update decision tree +

FlareGun event-routing decision tree +

Startup, shutdown, and cleanup safety tree +

---

11. Bottom line

> **Best concise description:** CPR+ is an endpoint network-state remediation and telemetry framework. It reduces support ambiguity by letting the endpoint determine its real access condition, take bounded corrective action, and leave behind readable evidence for IT.

Its strongest use cases are corporate Wi-Fi / ISE posture recovery, captive portal handling, VPN hairpin prevention, wake/network-change cleanup, endpoint-side event detection through Haystack, and safer field rollout through config sync plus self-update.

Generated as a CPR+ capability inventory draft. Logo is loaded from the docs folder next to this file. This document intentionally avoids sales claims and sticks to observed/reported framework behavior.